

Axel VandenHeuvel

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EDUCATION

University of Colorado Boulder

Boulder, CO

Bachelor of Science in Computer Science

Expected May 2026

- GPA: 3.64/4.0 - Computer Science GPA: 3.96/4.0
- Relevant Coursework: Data Structures, Algorithms, Software Development, Databases, Data Science, Cyber Security, Artificial Intelligence, Machine Learning, Object Oriented Design, AI Tooling in Software Dev.

EXPERIENCE

VLMs for Deep Space Spacecraft

August 2025 - Present

NASA JPL Capstone | C++, VLMs, Linux

Boulder, CO

- Collaborating with NASA's Jet Propulsion Laboratory to develop an open source software library to run and evaluate VLMs for deep-space spacecraft.
- Building a custom runtime and evaluation pipeline in C++ to deploy models (e.g., gemma3) on low-power systems such as Raspberry Pi-class devices.

Cloud Solutions / Azure Intern

June 2025 - August 2025

Tenet Healthcare

Dallas, TX

- Reclaimed unused Microsoft 365 and Azure licensing through automated audit and analysis pipelines, contributing to more than \$500K in projected annual cost savings
- Integrated the DocuSign API to automate reminder and compliance notification workflows, improving visibility into outstanding documents and reducing internal processing delays.
- Built PowerShell automation to detect misconfigurations and manage entitlement workflows across 10,000+ users.

PROJECTS

Local Speech-to-Text Desktop App | Rust, Tauri, whisper.cpp, llama.cpp

April 2026

- Built and shipped a macOS speech-to-text app with fully local AI - no cloud, no API keys.
- Bundled whisper.cpp for transcription and a statically-compiled llama.cpp server running Qwen 2.5 1.5B for grammar refinement, with Metal GPU acceleration.
- [GitHub Link](#), [Website](#)

GraphRAG Knowledge System | Python, FastAPI, TypeScript

Hackathon Project March 2026

- Co-built a full-stack app that ingests notes (MD/PDF/Notion), extracts concepts with an LLM, and renders them as an interactive 3D force-directed graph.
- Designed a hybrid retrieval pipeline combining LanceDB vector search with Kuzu weighted co-occurrence graph traversal, streaming the traversal plan to animate query execution live.
- [GitHub Link](#)

Lift-Tracker iOS App | Swift, Claude

March 2026

- Architected and deployed an iOS application for tracking weightlifting progression, utilizing AI coding assistants to accelerate the development lifecycle from concept to MVP.
- Designed a minimalist user interface, prioritizing an intuitive, distraction-free UX for mid-workout logging.
- [App Store Link](#), [GitHub Link](#)

Fact-Checker Chrome Extension | Python, JavaScript, Flask, OpenAI

Hackathon Project March 2025

- Built a Chrome extension that analyzes online claims and returns a fact-check score + source-backed summary in one click.
- Implemented retrieval and reasoning using OpenAI models + contextual search results.
- [DevPost Link](#), [GitHub Link](#)

TECHNICAL SKILLS & INTERESTS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: Flask, NumPy, pandas

Developer Tools: Git, Claude Code, Codex, Docker

Other: Ollama / llama.cpp, Raspberry Pi, Agentic AI, n8n

Interests: Weight-Lifting, Mechanical Keyboards, Astronomy, Hiking